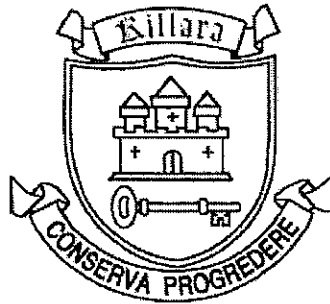


KILLARA HIGH SCHOOL



2009

HALF-YEARLY HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time- 5 minutes
- Working Time – 3 hours
- Write using a blue or black pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Untidy work or work with poor logic may be penalised.

Total marks (120)

- Attempt Questions 1-8
- All questions are of equal value
- Start a fresh booklet of paper for each question.
- Put your name and the question number at the top of booklet.

Total marks - 120

Attempt questions 1 – 8

All questions are of equal value

Answer each question on a separate writing booklet.

Question 1 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Sketch the graph of $y = (2x + 1)(x + 1)$ showing clearly the intercepts on the coordinate axes and the coordinates of any turning points. 2

- (b) Use the graph in part (a) to sketch the following graphs:
(equations of asymptotes, turning points and intercepts must be shown)

(i) $y = \frac{1}{(2x+1)(x+1)}$ 2

(ii) $y = |(2x+1)(x+1)|$ 2

(iii) $y = \sqrt{(2x+1)(x+1)}$ 2

(iv) $y = \ln\{(2x+1)(x+1)\}$ 3

- (c) Find the equation of the tangent to the curve $y = \ln\{(2x+1)(x+1)\}$ at the origin. 4

Hence find the values of the real number k such that exactly one of the solutions of the equation $\ln\{(2x+1)(x+1)\} = kx$ is a positive number.

- (a) $-3 + 4i$ has two square roots z_1 and z_2 . Find z_1 and z_2 in the form $a + ib$. 2

Show the points representing $-3 + 4i$, z_1 and z_2 on an Argand diagram. Show that These three points are vertices of a right-angled triangle. 2

- (b) The complex number z is represented by the point P on an Argand diagram. Indicate clearly on a single diagram the locus of P in each of the following cases. 4

(i) $|z - 4| = |z + 2i|$

(ii) $\arg(z - 3) = \frac{\pi}{4}$

Show that there is a point representing the complex number of the form ib , where b is real, which lies on both loci.

- (c) The equation $z^2 + (1 - 2i)z - (7 + i) = 0$ has roots α and β . 1
Find the monic equation with numerical coefficients α and $\beta - i$.
Hence find the values of α and β .

- (d) On an Argand diagram the point A represents the real number 1. O is the origin and the point P represents the complex number z which satisfies the condition $\arg(z - 1) = 2\arg(z)$.

(i) Show this information on a diagram and deduce that the triangle OAP is isosceles. 3

(ii) Deduce that the locus of P is a circle and show this circle on your diagram. 1

(iii) Find z in modulus/argument form if z also satisfies the condition $|z| = |z - 1|$. 2

- (a) The equation $x^3 + 2x - 1 = 0$ has roots α , β and γ . In each of the following cases, find the equations with the roots as stated.
- | | | |
|-------|---|---|
| (i) | $-\alpha, -\beta, -\gamma$. | 1 |
| (ii) | $\alpha, -\alpha, \beta, -\beta, \gamma, -\gamma$. | 2 |
| (iii) | $\alpha^2, \beta^2, \gamma^2$ | 2 |

- (b) (i) Prove that for any polynomial $P(x)$, if k is a zero multiplicity 2, then k is also a zero of $P'(x)$. 2

- (ii) Show that $x = 1$ is a double root of the equation: 2

$$x^{2n} - nx^{n+1} + nx^{n-1} - 1 = 0.$$

- (iii) Find all the roots of $3x^3 - 26x^2 + 52x - 24 = 0$, given that the roots are in geometric progression. 3

- (c) On the same set of axes, sketch the graphs of 3

$$y = \frac{1}{2}x + 1 \text{ and } y = |1 - x|.$$

Hence, or otherwise determine the values of x for which $\frac{1}{2}x + 1 \geq |1 - x|$.

Question 4 (15 marks) Use a SEPARATE writing booklet**Marks**

(a) Find:

(i) $\int \frac{e^x}{(1 - e^x)^2} dx$ 2

(ii) $\int_1^6 x\sqrt{6-x} \, dx$ 3

(iii) Use the substitution $x = \sin^2 \theta$ to evaluate $\int_0^{\frac{1}{2}} \frac{\sqrt{x}}{(1-x)^{3/2}} dx$ 4

(b) (i) By considering the sum of the terms of an arithmetic series, show that 2
 $(1 + 2 + 3 + \dots + n)^2 = \frac{1}{4}n^2(n+1)^2$

(ii) By principle of mathematical induction, prove that 4

$$1^3 + 2^3 + 3^3 + \dots + n^3 = (1 + 2 + 3 + \dots + n)^2$$

Question 5 (15 marks) Use a SEPARATE writing booklet

Marks

(a) (i) Solve $\tan 4\theta = 1$, $0 \leq \theta \leq \pi$. 2

(ii) Prove that $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$. 3

(iii) Using (i) and (ii) solve $z^4 + 4z^3 - 6z^2 - 4z + 1 = 0$ 3

(iv) Hence, find:

(α) $\tan \frac{\pi}{16} + \tan \frac{5\pi}{16} + \tan \frac{9\pi}{16} + \tan \frac{13\pi}{16}$ 1

(β) $\tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16}$ 2

(d) (i) If $2\sin 2x + \cos 2x = k$, show that: 2

$$(1+k) \tan^2 x - 4 \tan x - 1 + k = 0.$$

(ii) Also, show that if $\tan x_1$ and $\tan x_2$ are the roots of this quadratic equation in $\tan x$, then $\tan(x_1 + x_2) = 2$. 2

Question 6 (15 marks) Use a SEPARATE writing booklet

Marks

(a) (i) Sketch the curve $y = 2x - \frac{1}{x-2}$ 2

(ii) Evaluate $\int_{-1}^1 2x - \frac{1}{x-2} dx$. 3
Express your answer in the form $\ln k$, where k is a real constant.

(b) (i) Explain why the parametric coordinates of the ellipse $4x^2 + 9y^2 = 36$ is $(3 \cos \theta, 2 \sin \theta)$ 1

(ii) What are the parametric coordinates of the ellipse of the equation 2

$$\frac{(x+1)^2}{4} + \frac{(y-3)^2}{2} = 1?$$

(c) The hyperbola $\mathcal{H}$ has Cartesian equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $P(x_1, y_1)$ is a point on $\mathcal{H}$.
The tangent to the hyperbola through P cuts the directrices at T and T' .

(i) Find the equation of the tangent to the hyperbola at the point P . 2

(ii) Find the coordinates of T . 2

(iii) Show that PT subtends a right angle at the focus S . 3

Question 7 (15 marks) Use a SEPARATE writing booklet

Marks

(a) For the ellipse $9x^2 + 16y^2 = 144$, find the:

(i) eccentricity 2

(ii) co-ordinates of the foci 1

(iii) equations of the directrices. 1

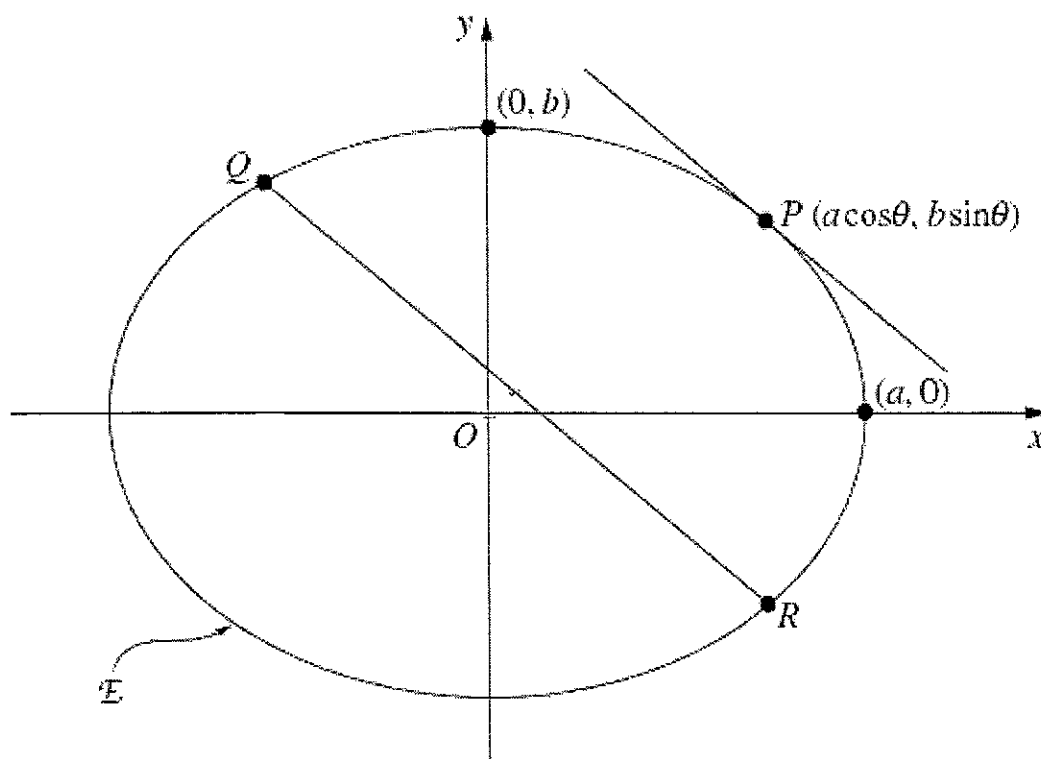
Hence,

(iv) sketch the ellipse showing the vertices, foci and directrices. 2

(v) Find the length of the latus rectum of the ellipse given above. 2

Question 7 continues on page 9

(b)



Consider the ellipse $\mathcal{E}$ with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and the points $P(a \cos \theta, b \sin \theta)$, $Q(a \cos(\theta + \varphi), b \sin(\theta + \varphi))$ and $R(a \cos(\theta - \varphi), b \sin(\theta - \varphi))$ on $\mathcal{E}$.

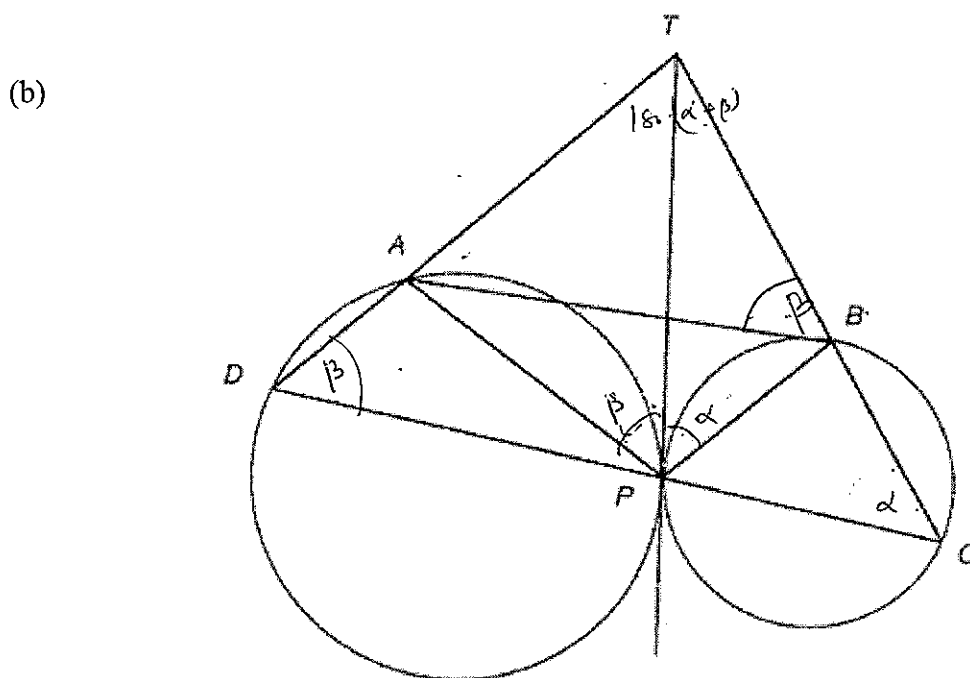
- (i) Show that the equation of the tangent to $\mathcal{E}$ at the point P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. 2
- (ii) Show that the chord QR is parallel to the tangent at P . 2
- (iii) Deduce that OP bisects the chord QR . 3

(a) The quadratic equation $x^2 - x + k = 0$, where k is a real number, has two distinct positive real roots α and β .

(i) Show that $0 < k < \frac{1}{4}$ 2

(ii) Show that $\alpha^2 + \beta^2 = 1 - 2k$ and deduce that $\alpha^2 + \beta^2 > \frac{1}{2}$. 2

(iii) Show that $\frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8$. 2



Circles APD and BPC touch at P . D , P and C are collinear.

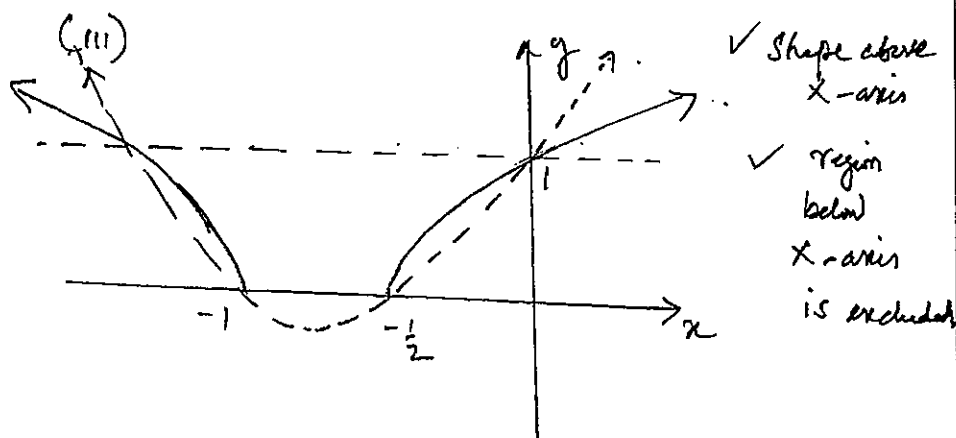
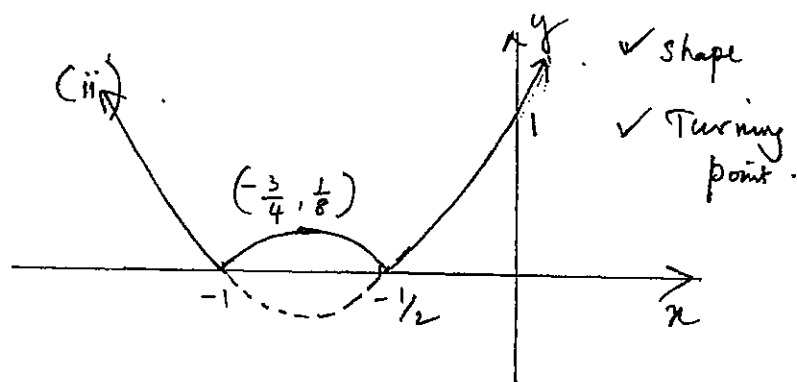
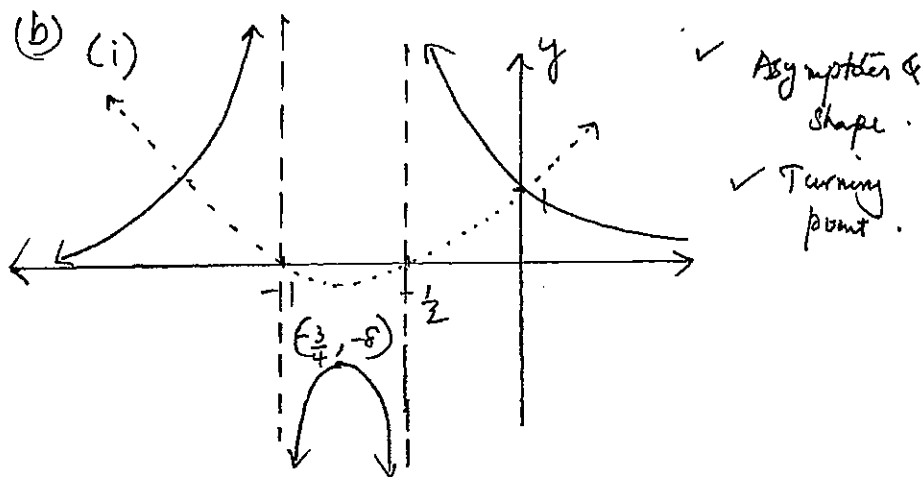
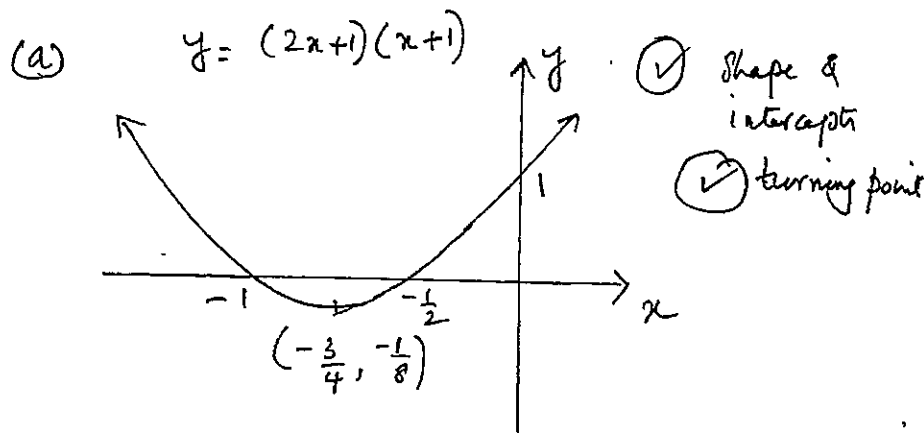
TP is the common tangent at P . TC cuts circle BPC in B while TD cuts circle APD in A .

- (i) Copy the diagram.
- (ii) Show that $ATBP$ is a cyclic quadrilateral. 4
- (iii) Show that $ABCD$ is a cyclic quadrilateral. 3

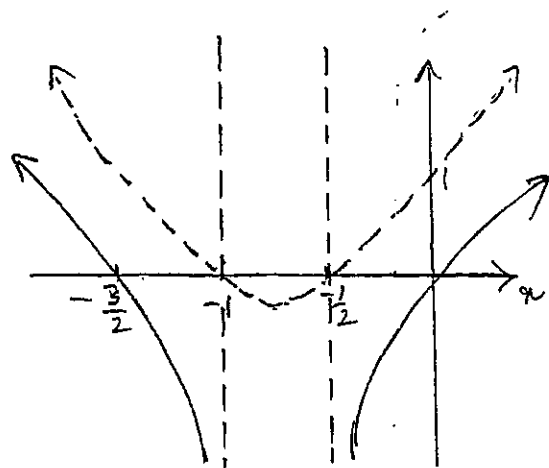
End of paper

Mathematics Extension 2 - Half-Yearly 2009

Questions 1



(iv) $y = \ln\{(2x+1)(x+1)\}$



Asymptotes ✓
Intercepts, $-1 < x < -\frac{1}{2}$ ignored
shape ✓

(c) $y = \ln(2x+1)(x+1)$

$= \ln(2x+1) + \ln(x+1)$

$\frac{dy}{dx} = \frac{2}{2x+1} + \frac{1}{x+1}$

$= \frac{2(x+1) + (2x+1)}{(2x+1)(x+1)}$

$= \frac{4x+3}{(2x+1)(x+1)}$

$\frac{dy}{dx}$ at $x=0$

is $\frac{3}{1} = 3$

Equation of tangent at (0,0)

is $y = 3x$

Exactly one positive solution for $0 < k < 3$

Question 2.

(a) $-3+4i = (a+ib)^2$

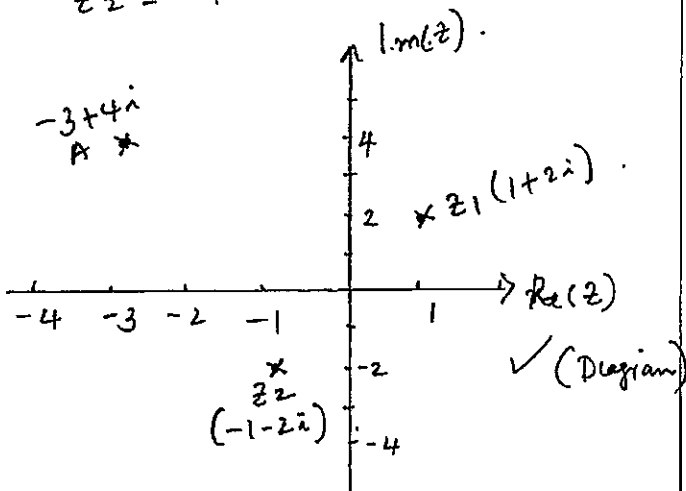
$$\left. \begin{aligned} a^2 - b^2 &= -3 \\ 2ab &= 4 \\ \therefore ab &= 2 \end{aligned} \right\} \checkmark$$

factoring $2i, 1$.

$$\therefore -3+4i = (1+2i)^2$$

$$\therefore \sqrt{-3+4i} = \pm (1+2i)$$

$$\left. \begin{aligned} z_1 &= 1+2i \\ z_2 &= -1-2i \end{aligned} \right\} \checkmark \quad (2)$$



$$\text{gradient of } Az_1 = \frac{4}{-2} = -2$$

$$\text{grad. of } z_1 z_2 = \frac{4}{2} = 2$$

$$\text{grad. of } Az_2 = \frac{2}{-4} = -\frac{1}{2}$$

$$m_{z_1 z_2} \times m_{Az_1} = 2 \times -\frac{1}{2} = -1$$

$\therefore \triangle Az_1 z_2$ is right-angled. $\checkmark$

(b)(i) $|z-4| = |z+2i|$

$$|z-(4+0i)| = |z-(0-2i)|$$

(2)

The locus is the perpendicular bisector of the line joining

A(4,0) and B(0,-2)

$$m_{AB} = \frac{2}{4} = \frac{1}{2} \text{ Mid-point } (2, -1)$$

equation of the perpendicular bisector is

$$y+1 = -2(x+2) \quad \checkmark \quad (1)$$

$2x+y-3=0$ is the locus of z.

(ii) $\arg(z - (-3+0i)) = \frac{\pi}{4}$

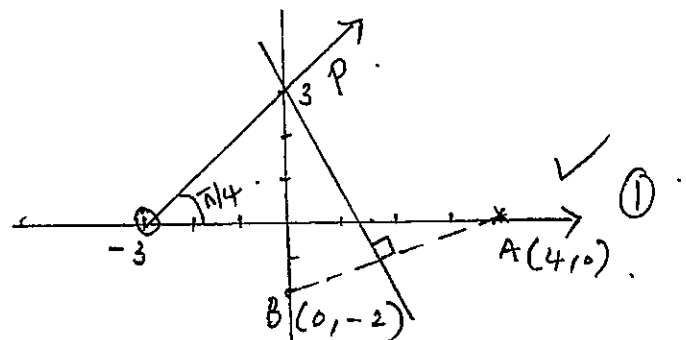
The line from P to (-3,0)

makes an angle of $\frac{\pi}{4}$ $\therefore m=1$

$\therefore$ equation of the line is

$$y = 1(x+3) \quad (1)$$

$$x-y+3=0 \quad \checkmark$$



P lies both on $2x+y-3=0$

and $x-y+3=0$.

$$\therefore P(0,3) \quad \checkmark \quad (1)$$

P is complex number $3i$

(9). $P(z) = z^2 + (1-2i)z - (7+i) = 0$.
has roots α and β .

$\therefore P(z+i)$ has roots $\alpha-i, \beta-i$.

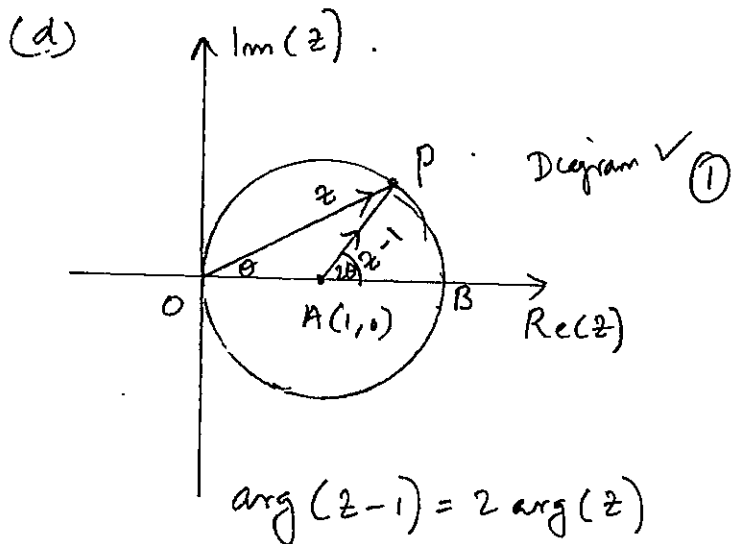
$$P(z+i) = (z+i)^2 + (1-2i)(z+i) - (7+i) = 0.$$

$$= z^2 + 2iz - 1 + z + i - 2iz + 2 - 7 - i = 0.$$

$$z^2 + z - 6 = 0. \quad \checkmark \quad (1)$$

$\therefore P(z+i) = 0$ has roots $-3, 2$.

α and β are $-3+i, 2+i$.



Let $\arg(z) = \theta \therefore \angle POA = \theta$

$\therefore \arg(z-1) = 2\arg(z) = 2\theta. \quad \checkmark$

i.e. $\angle PAB = 2\theta$.

$\therefore \angle OPA = \theta$. (exterior angle = sum of the opposite interior angles.)

$\therefore \angle OPA = \angle POA = \theta$.

(3)

$\therefore AO = AP$ (Sides opposite equal angles are equal)

$\therefore \triangle OAP$ is isosceles. $\checkmark$ (2)

(ii) $PA = 1$.

i.e. $|z-1| = 1$

$\therefore |z - (1+0i)| = 1$.

$\therefore P$ moves on a circle with centre $(1,0)$ and radius 1. $\checkmark$ (1)

(iii) $|z| = |z-1|$

$\therefore \triangle OAP$ is equilateral with side 1 unit.

$\therefore |z| = 1$ and $\arg(z) = \frac{\pi}{3}. \quad \checkmark$ (2)

i.e. $z = 1 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \checkmark$

Question 3.

(a) (i) $x^3 + 2x - 1 = 0$ has roots α, β, γ .

To find equation with roots $-\alpha, -\beta$ and $-\gamma$, replace x by $-x$.

$$\therefore (-x)^3 + 2(-x) - 1 = 0 \quad \checkmark \quad (1)$$

ie $x^3 + 2x + 1 = 0$ has roots $-\alpha, -\beta, -\gamma$.

(ii) $x^3 + 2x - 1 = 0$ has roots α, β, γ .
 $x^3 + 2x + 1 = 0$ has roots $-\alpha, -\beta, -\gamma$ } $\checkmark$

$$\therefore (x^3 + 2x - 1)(x^3 + 2x + 1) = 0$$

has roots $\pm\alpha, \pm\beta, \pm\gamma$.

$$(x^3 + 2x)^2 - 1 = 0 \quad \checkmark \quad (2)$$

$$\underline{\underline{x^6 + 4x^4 + 4x^2 - 1 = 0}}$$

(iii) To find equation with roots $\alpha^2, \beta^2, \gamma^2$, replace x by $\sqrt{x}$.

$$(\sqrt{x})^3 + 2\sqrt{x} - 1 = 0$$

$$x^{3/2} + 2x^{1/2} - 1 = 0$$

$$x^{1/2} (x+2) = 1$$

$$x+2 = x^{-1/2}$$

Squaring,

$$x^2 + 4x + 4 = \frac{1}{x}$$

(4)

$x^3 + 4x^2 + 4x - 1 = 0$ is the required equation.

(b) (i) $P(x) = (x-k)^2 Q(x) \quad \checkmark$

$$P'(x) = 2(x-k)Q(x) + (x-k)^2 Q'(x)$$

$$P'(k) = 2(k-k)Q(k) + (k-k)^2 Q'(k) = 0 \quad \checkmark$$

ie. k is also a root of $P'(x) = 0$.

(ii) Let $P(x) = x^{2n} - nx^{n+1} + nx^{n-1} - 1$

$$P(1) = 1 - n + n - 1 = 0$$

$\Rightarrow x=1$ is a root of $P(x) = 0 \quad \checkmark$

$$P'(x) = 2nx^{2n-1} - n(n+1)x^n + n(n-1)x^{n-2}$$

$$P'(1) = 2n - n(n+1) + n(n-1) = 2n - n^2 - n + n^2 - n = 0$$

$\Rightarrow x=1$ is a root of $P'(x) = 0$ and $x=1$ is a double root.

(iii) Let the roots be α, β, γ .

The roots are in G.P.

$$\therefore \beta^2 = \alpha\gamma \quad (1)$$

$$3x^3 - 26x^2 + 52x - 24 = 0$$

$$\alpha + \beta + \gamma = \frac{26}{3} \quad (2)$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{52}{3} \quad (3)$$

$$\alpha\beta\gamma = 8 \quad (4)$$

From (i) and (4),

$$\beta^3 = 8 \quad \therefore \beta = 2 \quad \checkmark$$

From (2) and (4)

$$\alpha + r = \frac{26}{3} - 2 = \frac{20}{3}$$

$$\alpha r = 4$$

$$\therefore \alpha + \frac{4}{\alpha} = \frac{20}{3}$$

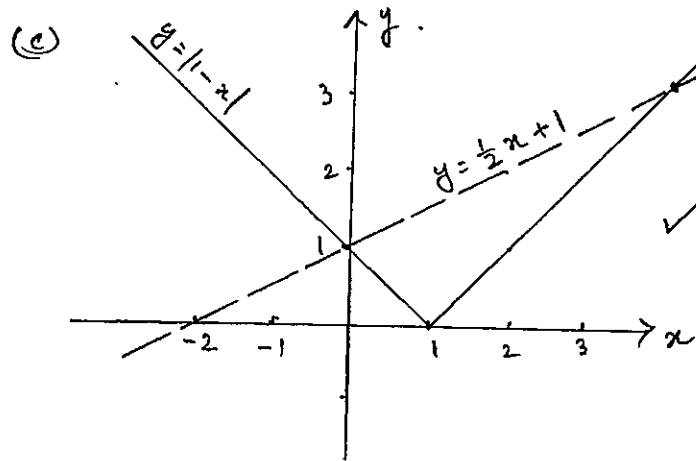
$$3\alpha^2 - 20\alpha + 12 = 0$$

$$(3\alpha - 2)(\alpha - 6) = 0 \quad \boxed{3}$$

$$\alpha = \frac{2}{3} \text{ or } \alpha = 6 \quad \checkmark$$

$$r = 6 \quad r = \frac{2}{3}$$

$\therefore$ Roots are $\frac{2}{3}$, 2, and 6.



The graphs intersect when

$$x - 1 = \frac{1}{2}x + 1$$

$$\therefore x = 4 \quad \checkmark$$

From graph, $\boxed{3}$

$$\frac{1}{2}x + 1 \geq |1 - x| \text{ when}$$

$$0 \leq x \leq 4 \quad \checkmark$$

(5)

Question 4

(a) (i) $\int \frac{e^x}{(1-e^x)^2} dx$

$$\text{let } 1 - e^x = u$$

$$-e^x dx = du \quad \checkmark$$

$$\therefore -\int \frac{1}{u^2} du$$

$$= -\frac{u^{-1}}{-1} + C$$

$$= \frac{1}{1-e^x} + C \quad \checkmark \quad \textcircled{2}$$

(ii) $\int_1^6 x \sqrt{6-x} dx$

$$\text{let } 6-x = u \quad \therefore x = 6-u$$

$$\therefore dx = -du$$

$$x=1, u=5$$

$$x=6, u=0 \quad \checkmark \quad \textcircled{1}$$

$$\therefore -\int_5^0 (6-u) \sqrt{u} du$$

$$\int_0^5 (6-u) \sqrt{u} du$$

$$= \int_0^5 6u^{\frac{1}{2}} - u^{\frac{3}{2}} du \quad \checkmark \quad \textcircled{1}$$

$$= \left[6 \cdot \frac{2}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} \right]_0^5$$

$$= \left[4u\sqrt{u} - \frac{2}{5} u^2 \sqrt{u} \right]_0^5$$

$$= \left(4 \times 5\sqrt{5} - \frac{2}{5} \times 25\sqrt{5} \right) - 0$$

$$= \underline{10\sqrt{5}} \quad \checkmark \quad \textcircled{1}$$

$$(iii) \int_0^{1/2} \frac{\sqrt{x}}{(1-x)^{3/2}} dx.$$

$$\left\{ \begin{array}{l} \text{let } x = \sin^2 \theta. \quad \therefore dx = 2 \sin \theta \cos \theta d\theta \\ x=0, \theta=0 \\ x=1/2, \theta=\pi/4. \end{array} \right. \quad \checkmark \quad (1)$$

$$\therefore \int_0^{\pi/4} \frac{\sin \theta}{(1-\sin^2 \theta)^{3/2}} \cdot 2 \sin \theta \cos \theta d\theta.$$

$$= \int_0^{\pi/4} \frac{\sin \theta}{\cos^3 \theta} \cdot 2 \sin \theta \cos \theta d\theta.$$

$$= 2 \int_0^{\pi/4} \tan^2 \theta d\theta. \quad \checkmark \quad (1)$$

$$= 2 \int_0^{\pi/4} (\sec^2 \theta - 1) d\theta. \quad \checkmark \quad (1)$$

$$= 2 \left[\tan \theta - \theta \right]_0^{\pi/4}$$

$$= 2 \left(1 - \frac{\pi}{4} \right) - 0.$$

$$= 2 \left(1 - \frac{\pi}{4} \right) \quad \checkmark \quad (1)$$

(b) (i) $1+2+3+\dots+n$ A.P with
 $a=1, d=1, l=n.$

$$\therefore \text{Sum of series} = \frac{n}{2} (1+n) \quad \checkmark$$

$$\therefore (1+2+3+\dots+n)^2$$

$$= \left[\frac{n}{2} (1+n) \right]^2$$

$$= \frac{1}{4} (n^2) (n+1)^2 \quad \checkmark \quad (2)$$

(ii) Let S_n be the sum of
 n terms in the series
 $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4} (n^2) (n+1)^2$

Step 1 When $n=1$.

$$S_1 = 1^3 = \frac{1}{4} \times 1^2 (1+1)^2$$

$$= 1$$

$$\text{LHS} = \text{RHS}.$$

$\therefore S_1$ is true. ✓

Step 2 Assume S_k is true.

$$\text{i.e. } 1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4} k^2 (k+1)^2$$

Step 3 Prove S_{k+1} is true.

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 =$$

$$\frac{1}{4} k^2 (k+1)^2 + (k+1)^3$$

$$= \frac{1}{4} (k+1)^2 [k^2 + 4(k+1)]$$

$$= \frac{1}{4} (k+1)^2 [k^2 + 4k + 4] \quad (4)$$

$$= \frac{1}{4} (k+1)^2 (k+2)^2 \quad \checkmark$$

$\therefore S_{k+1}$ is true, if S_k is true.

In step 1, S_1 is true.

$\therefore$ By principle of mathematical induction, S_n is true for

$n=2, 3, \dots$ all integral values of n . Using result (i). ✓

$$1^3 + 2^3 + 3^3 + \dots + n^3 = (1+2+3+\dots+n)^2$$

Question 5

(a) (i) $\tan 4\theta = 1$. $\frac{s}{c} = \frac{1}{1}$ ✓
 $\therefore 4\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$ ✓
 $\therefore \theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$ ✓ 2

(ii) Consider $(\cos\theta + i\sin\theta)^4$

Let $c = \cos\theta$, $s = \sin\theta$.

By De-Moivre theorem,

$$\cos 4\theta + i\sin 4\theta = (\cos\theta + i\sin\theta)^4$$

$$1, \quad 4, \quad -6, \quad -4, \quad 1$$

$$\cos 4\theta + i\sin 4\theta = (c + is)^4$$

$$= c^4 + 4c^3s - 6c^2s^2 - 4cs^3 + s^4$$

$$\therefore \cos 4\theta = c^4 - 6c^2s^2 + s^4$$

$$\sin 4\theta = 4c^3s - 4cs^3$$

$$\therefore \tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta}$$

$$= \frac{4c^3s - 4cs^3}{c^4 - 6c^2s^2 + s^4}$$
 3

Dividing by c^4 on RHS

$$\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}$$
 ✓

(iii) $z^4 + 4z^3 - 6z^2 - 4z + 1 = 0$.

from (i), $\tan 4\theta = 1$.

Using (ii), $\therefore$

$$\therefore \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta} = 1$$
 ✓

Let $z = \tan\theta$.

$$\therefore 4z - 4z^3 = 1 - 6z^2 + z^4$$

$$\therefore z^4 - 4z^3 - 6z^2 - 4z + 1 = 0$$

Using (i),

$$\theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$$
 ✓ 3

$$\therefore z = \tan\theta =$$

$$\tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$$
 ✓

(iv) Sum of

$$(x) z^4 + 4z^3 - 6z^2 - 4z + 1 = 0$$

$$\text{Sum of roots} = \frac{-b}{a} = -4$$

$$\therefore \tan \frac{\pi}{16} + \tan \frac{5\pi}{16} + \tan \frac{9\pi}{16} + \tan \frac{13\pi}{16}$$

$$= -4$$
 1

(p) $(\alpha + \beta + \gamma + \delta)^2 = \sum \alpha^2 + 2\sum \alpha\beta$ ✓

$$(-4)^2 = \sum \alpha^2 + 2x - 6$$
 2

$$\therefore \sum \alpha^2 = 16 + 12 = 28$$

$$\therefore \tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16}$$

$$= 28$$

(b) Let $t = \tan x$.

(i) Then $2\sin 2x + \cos 2x = k$

$$2 \times \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = k \quad \checkmark$$

$$4t + 1 - t^2 = k(1+t^2)$$

$$4t + 1 - t^2 = k + kt^2$$

$$t^2(k+1) - 4t + k-1 = 0 \quad \checkmark$$

i.e. $(k+1)\tan^2 x - 4\tan x + k-1 = 0$

(ii) $\tan x_1$ and $\tan x_2$ are roots of this equation.

$\therefore$ Sum of roots, product of roots

$$\tan x_1 + \tan x_2 = \frac{4}{k+1} \quad \checkmark$$

$$\tan x_1 \cdot \tan x_2 = \frac{k-1}{k+1} \quad \checkmark$$

$$\tan(x_1 + x_2) = \frac{\tan x_1 + \tan x_2}{1 - \tan x_1 \cdot \tan x_2}$$

$$= \frac{\frac{4}{k+1}}{1 - \frac{k-1}{k+1}} = \frac{4}{k+1 - k+1}$$

$$= \frac{4}{2} = 2 \quad \checkmark$$

Question 6.

(a) (i) $y = 2x - \frac{1}{x-2}$

$\therefore$ asymptotes $x=2$.

Oblique asymptote $y = 2x$.

y-intercept

Let $x=0$. $\therefore y = \frac{1}{2}$.

x-intercepts

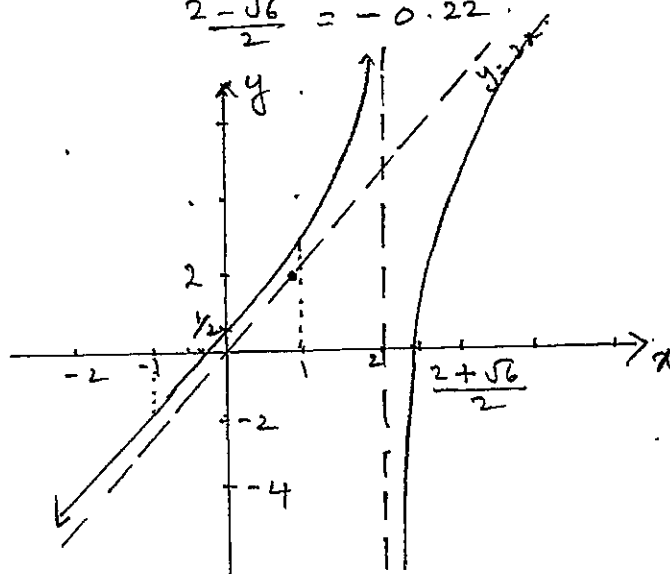
Let $y=0$.

$$\therefore 2x = \frac{1}{x-2}$$

$$2x^2 - 4x - 1 = 0.$$

$$x = \frac{4 \pm \sqrt{24}}{4} = \frac{2 \pm \sqrt{6}}{2} \approx 2.22$$

$$\frac{2 - \sqrt{6}}{2} = -0.22$$

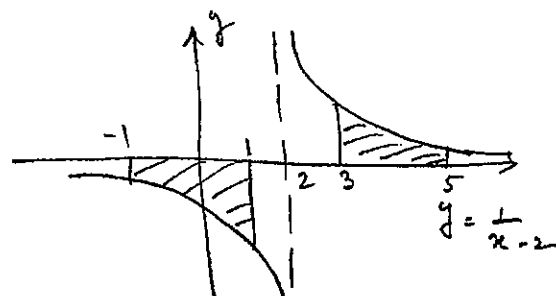


$\checkmark$ Intercepts, asymptotes.

$\checkmark$ Shape.

(ii) $\int_{-1}^1 2x - \frac{1}{x-2} dx$

$$= \int_{-1}^1 2x dx - \int_{-1}^1 \frac{1}{x-2} dx \quad \checkmark$$



$$= \left[x^2 \right]_{-1}^1 - \left[\ln(x-2) \right]_{-1}^1$$

When $x=1$ or -1 ,

$x-2$ is negative.

$\therefore \ln e_{-(x-2)}$ doesn't exist.

But from the graph of

$y = \frac{1}{x-2}$, it can be

observed that

$\int_{-1}^1 \frac{1}{x-2} dx$ is symmetrical
to $\int_3^5 \frac{1}{x-2} dx$.

$$\left[x^2 \right]_{-1}^1 - \left[\ln(x-2) \right]_3^5$$

$$= 0 - [\ln(3) - \ln(1)]$$

$$= 0 - \ln 3 + 0 = -\ln 3.$$

But $\int_{-1}^1 \frac{1}{x-2} dx$ is below

x -axis. $\therefore -(-\ln 3) = \ln 3$.

$$\therefore \int_{-1}^1 2x - \frac{1}{x-2} dx = \underline{\underline{\ln 3}}$$

$$\therefore k = \underline{\underline{3}}$$

[3]

OR

$$\int_{-1}^1 2x - \frac{1}{x-2} dx$$

$$= \left[x^2 - \ln|x-2| \right]_{-1}^1$$

$$= (1 - \ln|-1|) - (1 - \ln|-3|)$$

$$= 1 - 0 - 1 + \ln 3$$

$$= \ln 3$$

$$\therefore k = 3$$

$$(b) (i) 4x^2 + 9y^2 = 36$$

(9)

$$x = 3 \cos \theta, \quad y = 2 \sin \theta$$

$$4(3 \cos \theta)^2 + 9(2 \sin \theta)^2$$

$$= 4 \times 9 \cos^2 \theta + 9 \times 4 \sin^2 \theta$$

$$= 36 (\cos^2 \theta + \sin^2 \theta) = 36 = \text{RHS.}$$

ie $(3 \cos \theta, 2 \sin \theta)$ satisfies

$$4x^2 + 9y^2 = 36$$

$\therefore (3 \cos \theta, 2 \sin \theta)$ is the

parametric representation of any

point on $4x^2 + 9y^2 = 36$ ✓ [1]

$$(ii) \frac{(x+1)^2}{4} + \frac{(y-3)^2}{2} = 1$$

$$a = 2 \quad b = \sqrt{2}$$

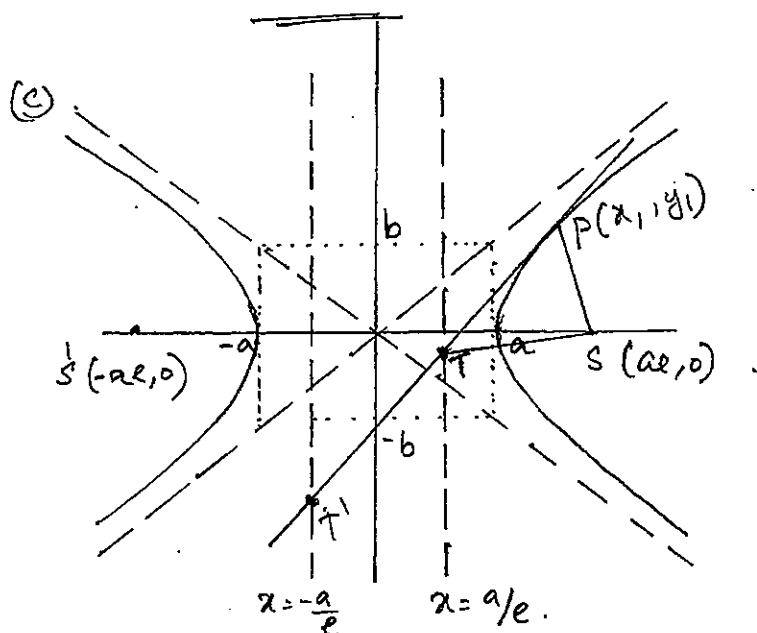
$$x+1 = 2 \cos \theta \quad \therefore x = 2 \cos \theta - 1$$

$$y-3 = \sqrt{2} \sin \theta \quad \therefore y = 3 + \sqrt{2} \sin \theta$$

$\therefore$ Parametric Coordinates are

$$(2 \cos \theta - 1, 3 + \sqrt{2} \sin \theta)$$

[2]



$$(i) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2yy'}{b^2} = 0$$

$$\frac{2yy'}{b^2} = \frac{2x}{a^2}$$

$$\therefore y' = \frac{2x}{a^2} \cdot \frac{b^2}{2y}$$

$$= \frac{b^2}{a^2} \frac{x}{y}$$

y' at $P(x_1, y_1)$ is

$$\frac{b^2}{a^2} \frac{x_1}{y_1} \checkmark$$

Equation of the tangent is

$$y - y_1 = \frac{b^2}{a^2} \frac{x_1}{y_1} (x - x_1)$$

$$a^2 y y_1 - a^2 y_1^2 = b^2 x x_1 - b^2 x_1^2$$

$$b^2 x x_1 - a^2 y y_1 = b^2 x_1^2 - a^2 y_1^2$$

Divide by $a^2 b^2$

$$\therefore \frac{x x_1}{a^2} - \frac{y y_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$$

(x_1, y_1) is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

$$\therefore \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$$

$\therefore$ Equation of the tangent at

$P(x_1, y_1)$ is

$\checkmark$ 2

$$\frac{x x_1}{a^2} - \frac{y y_1}{b^2} = 1$$

(ii)

Solve simultaneously equation of tangent and the directrix $x = \frac{a}{e}$ to get coordinates of T.

$$\frac{x x_1}{a^2} - \frac{y y_1}{b^2} = 1$$

$$x = \frac{a}{e}$$

$$\frac{x x_1}{e a^2} - \frac{y y_1}{b^2} = 1$$

$$\frac{y y_1}{b^2} = \frac{x_1}{a e} - 1 = \frac{x_1 - a e}{a e}$$

$$\therefore y = \frac{b^2 (x_1 - a e)}{y_1 a e} \checkmark$$

$$\therefore T \left[\frac{a}{e}, \frac{b^2 (x_1 - a e)}{a e y_1} \right] \checkmark$$

$$(iii) \quad m_{TS} = \frac{\frac{b^2 (x_1 - a e)}{a e y_1} - 0}{\frac{a}{e} - a e}$$

$$= \frac{b^2 (x_1 - a e)}{\frac{a y_1}{a - a e^2}}$$

$$= \frac{b^2 (x_1 - a e)}{a y_1 \times a (1 - e^2)}$$

$$= \frac{b^2 (x_1 - a e)}{a^2 y_1 (1 - e^2)} \checkmark$$

$$m_{PS} = \frac{y_1 - 0}{x_1 - ae}$$

$$m_{TS} \times m_{PS} = \frac{b^2(x_1 - ae)}{a^2 y_1 (1 - e^2)} \times \frac{y_1}{(x_1 - ae)}$$

$$= \frac{b^2}{a^2(1 - e^2)} = \frac{b^2}{-a^2(e^2 - 1)} \quad \checkmark$$

But $a^2(e^2 - 1) = b^2$, $e > 1$
for a hyperbola.

$$\therefore m_{TS} \times m_{PS} = \frac{b^2}{-b^2} = -1.$$

ie $TS \perp PS$. 3

ie PT subtends a right-angle at the focus.

Question 7

(a) $9x^2 + 16y^2 = 144$.

(i) $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

$a = 4$ $b = 3$. ✓

$\therefore b^2 = a^2(1 - e^2)$.

$9 = 16(1 - e^2)$.

$1 - e^2 = \frac{9}{16}$. (2)

$e^2 = 1 - \frac{9}{16} = \frac{7}{16}$

$\therefore e = \frac{\sqrt{7}}{4}$. ✓

(11)

(ii) foci are $(\pm ae, 0)$.

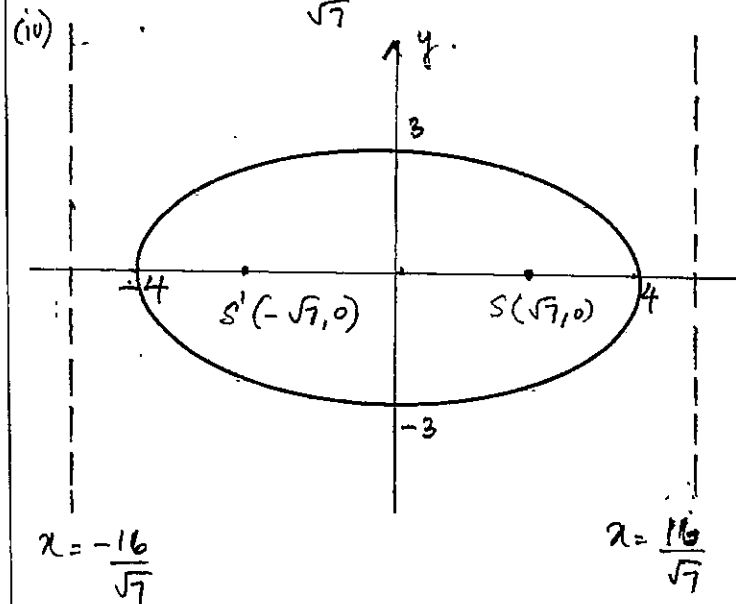
$(\pm 4 \times \frac{\sqrt{7}}{4}, 0)$

$S(\sqrt{7}, 0)$ $S'(-\sqrt{7}, 0)$. ✓ (1)

(iii) Directrices are

$x = \pm \frac{a}{e} = \pm \frac{4}{\frac{\sqrt{7}}{4}}$

$x = \pm \frac{16}{\sqrt{7}}$. ✓ (1)



✓ foci and vertices

(2)

✓ ... directrices shown and labelled.

(v)

$9x^2 + 16y^2 = 144$

The latus rectum (AA') is the chord through the focus parallel to the directrix. $9x^2 + 16y^2 = 144$. ✓

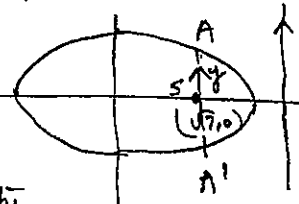
(2)

$63 + 16y^2 = 144$.

$16y^2 = 81$.

$y = \frac{9}{4}$

$\therefore AA' = 2 \times \frac{9}{4}$ ✓
 $= 4.5 \text{ units}$



(b) (i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$

$x = a \cos \theta \quad \frac{dx}{d\theta} = -a \sin \theta.$

$y = b \sin \theta \quad \frac{dy}{d\theta} = b \cos \theta.$

$\therefore \frac{dy}{dx} = -\frac{b}{a} \frac{\cos \theta}{\sin \theta} \quad \checkmark \quad \text{--- (1)}$

$\frac{dy}{dx}$ at

$\therefore$ Equation of the tangent is

$y - b \sin \theta = -\frac{b}{a} \frac{\cos \theta}{\sin \theta} (x - a \cos \theta).$

$a y \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta.$

$bx \cos \theta + a y \sin \theta = ab (\sin^2 \theta + \cos^2 \theta)$

$bx \cos \theta + a y \sin \theta = ab \times 1.$

Divide by ab ,

$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1. \quad \checkmark$

(ii) gradient of QR

$= \frac{b \sin(\theta + \phi) - b \sin(\theta - \phi)}{a \cos(\theta + \phi) - a \cos(\theta - \phi)}.$

$= \frac{b [\sin(\theta + \phi) - \sin(\theta - \phi)]}{a [\cos(\theta + \phi) - \cos(\theta - \phi)]}$

$\stackrel{(12)}{=} \frac{b \cdot 2 \cos \frac{\theta + \phi + \theta - \phi}{2} \sin \frac{\theta + \phi - \theta + \phi}{2}}{a \cdot 2 \sin \frac{\theta + \phi + \theta - \phi}{2} \sin \frac{\theta + \phi - \theta + \phi}{2}} \quad \checkmark$

$= -\frac{b}{a} \cdot \frac{\cos \theta \sin \phi}{\sin \theta \cos \phi}$

$= -\frac{b}{a} \frac{\cos \theta}{\sin \theta} \quad \checkmark$

Using (1) from (i),

$m_{QR} = m_{\text{tangent at } P}.$

$\therefore QR \parallel \text{tangent at } P.$

(iii) Let M be the mid-point of QR.

$M \left\{ \frac{a [\cos(\theta + \phi) + \cos(\theta - \phi)]}{2}, \frac{b [\sin(\theta + \phi) + \sin(\theta - \phi)]}{2} \right\}$

$= M \left\{ \frac{a \cdot 2 \cos \theta \cdot \cos \phi}{2}, \frac{b \cdot 2 \sin \theta \cos \phi}{2} \right\}$

$= M \{ a \cos \theta \cos \phi, b \sin \theta \cos \phi \} \quad \checkmark$

equation of OP is

$y = \frac{b \sin \theta}{a \cos \theta} x. \quad \checkmark$

prove that M satisfies equation of OP.

$b \sin \theta \cos \phi = \frac{b \sin \theta}{a \cos \theta} \cdot a \cos \theta \cos \phi \quad \checkmark$
 $= b \sin \theta \cos \phi \quad \text{TRUE.}$

ie OP bisects the chord QR.

[3]

Question 8.

(a) $x^2 - x + k = 0$.

Distinct roots α, β both positive

(i) $\alpha\beta > 0$ } $\therefore k > 0$. — ① ✓
 $\alpha\beta = k$ }

$\Delta > 0$.

$(-1)^2 - 4k > 0$

$1 - 4k > 0 \therefore 4k < 1 \Rightarrow k < \frac{1}{4}$ — ②

from ① & ②, ✓

$0 < k < \frac{1}{4}$. [2]

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$.

$= 1^2 - 2 \times k = 1 - 2k$. ✓

Using ② from part (i),

$\alpha^2 + \beta^2 > 1 - 2 \times \frac{1}{4}$ ✓

$\therefore \alpha^2 + \beta^2 > \frac{1}{2}$. [2]

(iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{1 - 2k}{k^2}$ ✓

$> \frac{1 - 2 \times \frac{1}{4}}{(\frac{1}{4})^2} = 8$. ✓

$\therefore \frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8$.

(iv) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$

$= 1 \times (1 - 2k - k)$ ✓

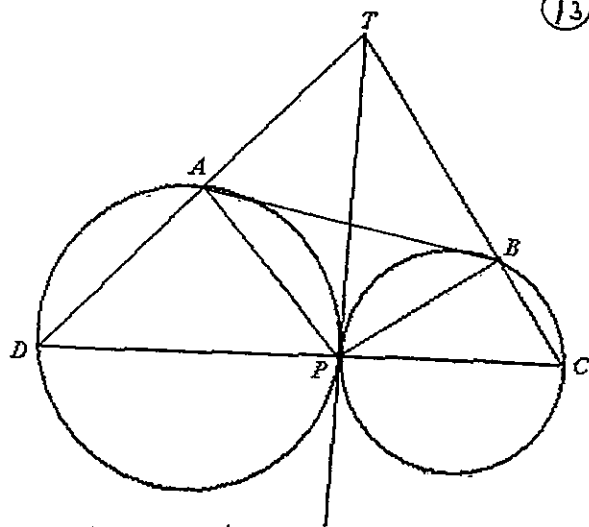
$= 1 - 3k$.

$> 1 - 3 \times \frac{1}{4}$ ✓

$> \frac{1}{4}$.

②

(i)



(ii) $\angle TPA = \angle BCP$ } Angle between — ①
 $\angle TPA = \angle ADP$ } tangent and chord

equal angle in the alternate segment. ✓

$\angle ATB + \angle ADP + \angle BCP = 180^\circ$ (sum of $\triangle TDC$) ✓

$\therefore \angle ATB + \angle TPA + \angle TPB = 180^\circ$ using ① ✓

$\therefore \angle ATB + \angle APB = 180^\circ$ ($\angle APB = \angle TPA + \angle TPB$) ✓

$\therefore$ ATBP is a cyclic quadrilateral.

(one pair of opposite $\angle$'s are supplementary). [4] ✓

(iii) $\angle TBA = \angle TPA$ ($\angle$'s at circumference standing on the $\widehat{AT}$ of circle ATBP). ✓

$\therefore \angle TBA = \angle ADP$ ($\angle TPA = \angle ADP$ proved in (ii)) ✓

$\therefore$ ABCD is cyclic (exterior angle equal to opposite interior angle). [3] ✓

— END —

